Design and Integration of a 2.45 GHz Communication and Antenna Measurement System

Antoni Gonzalez, Leo Joseph Alvelais, Paul Obrecht, Jack Marshall, Mark Negus, William Robert Pfaff

Department of Electrical Engineering, California Polytechnic State University, San Luis Obispo

EE541: Advanced Microwave Lab Spring 2026

I. Introduction

This project designed, built, and integrated a complete 2.45 GHz wireless link with an antenna measurement capability. The system transmits a baseband signal through an IQ modulator, amplifies and filters the resulting RF signal, radiates it from a transmit antenna, and recovers it at a receive chain consisting of a receive antenna, a low noise amplifier, a bandpass filter, and an IQ demodulator. A separate two-axis test stand was built to measure antenna radiation patterns.

The operating frequency of 2.45 GHz was chosen for its location in the ISM band. Each subsystem was designed and simulated independently, then fabricated and measured. The subsystems are a log-periodic dipole array antenna, an axial-mode helical antenna, a low noise amplifier, a fifth-order Chebyshev hairpin bandpass filter, a modulator and demodulator pair, and an antenna measurement test stand.

This report documents each subsystem, the link budget that ties them together, and the final integration test. The main result is that the assembled link demonstrated end-to-end functionality at 2.45 GHz. A secondary result is that accurate baseband capture at the receiver was not achieved, which is discussed in the results and conclusion.

II. System Overview

The signal path begins at the TRF3722 IQ modulator, which upconverts a baseband input to 2.45 GHz. The modulated signal is amplified by a power amplifier, filtered by a transmit bandpass filter, and radiated by a horn transmit antenna. The signal propagates across a 1 m test range and is received by the log-periodic dipole array. The received signal is amplified by the low noise amplifier, filtered by a receive bandpass filter, and downconverted by the TRF3711 IQ demodulator. The recovered baseband signal is sampled by the ADC of an STM32L4A6ZG microcontroller.

The antenna measurement test stand positions an antenna under test on a two-axis mount and records received power as a function of angle to produce a radiation pattern. The two bandpass filters and the two antenna designs were each developed so that either antenna could be used in the link or characterized on the test stand.

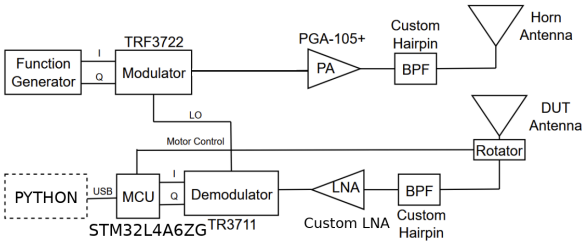


Figure 1. System block diagram.

III. Antennas

A. Log-Periodic Dipole Array

A log-periodic dipole array (LPDA) was designed to operate at 2.45 GHz and modeled in Ansys HFSS. The geometry was optimized for high gain and a favorable impedance match near the target frequency. The simulated model is shown in Fig. 2.

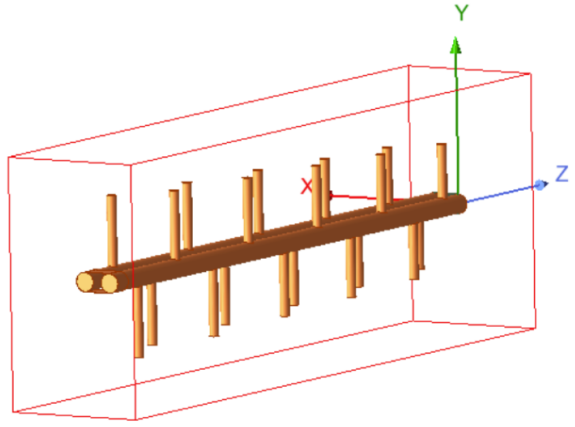


Figure 2. LPDA antenna model in the HFSS.

The simulated three-dimensional radiation pattern gives a maximum boresight gain of 11.8 dBi. The azimuthal gain cut is shown in Fig. 3.

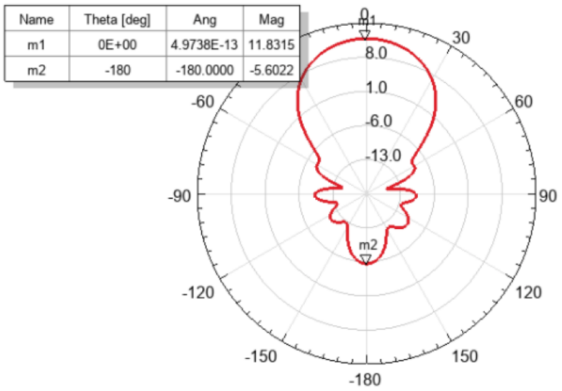


Figure 3. LPDA realized gain pattern, azimuthal cut.

The simulated S-parameter response shows the best impedance match near 2.6 GHz, where the return loss is about 25 dB. At the intended 2.45 GHz operating frequency, the return loss is approximately 11 dB, which indicates good matching near the design frequency. The return loss response is shown in Fig. 4.

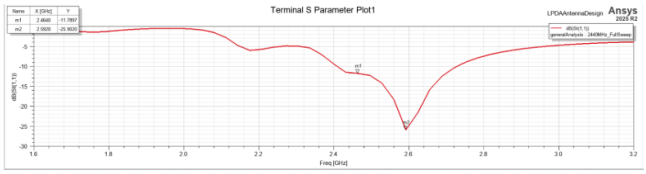


Figure 4. LPDA simulated return loss.

Fabrication of the LPDA is completed. A 3D-printed mounting structure was produced, and drill holes were placed along the boom to position the antenna elements. The electrical connections between the elements and the boom were then soldered. The LPDA was used as the receive antenna in the link budget. A final matching coefficient was measured to be near -8.5 dB at the target 2.45 GHz center design frequency. There also appeared to be a slight resonance upshift after fabrication, but the performance across the operating band was still acceptable and produced a meaningful response when scanning.

B. Axial-Mode Helical Antenna

An axial-mode helical antenna was also designed for 2.45 GHz and simulated in HFSS. The design consists of a helical wire structure, a copper ground plane approximately 122 mm in diameter, and an FR4 match track section for impedance matching at the target frequency. The model is shown in Fig. 5.

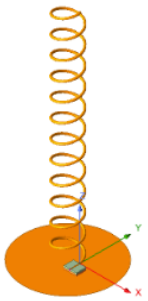


Figure 5. Axial-mode helical antenna model in HFSS.

The S-parameter sweep shows good input matching from approximately 2.21 GHz to 2.60 GHz, with return loss below 20 dB across that range. The strongest match is near 2.42 GHz, where the simulated return loss reaches approximately 29.4 dB. The return loss sweep is shown in Fig. 6.

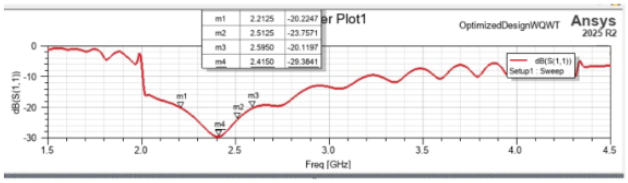


Figure 6. Helical antenna simulated return loss.

The simulated far-field results give a maximum boresight gain of about 12.3 dBi. The radiation pattern is shown in Fig. 7.

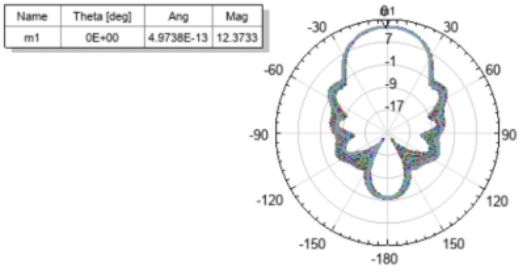


Figure 7. Helical antenna far-field radiation pattern.

The antenna also shows an axial-ratio beamwidth of approximately 68 degrees from boresight. The axial ratio is shown in Fig. 8.

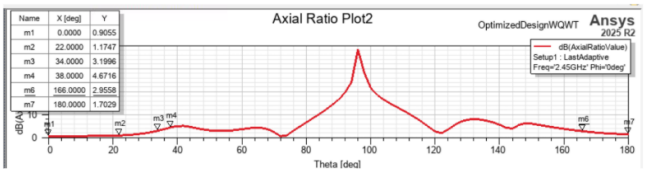


Figure 8. Helical antenna axial ratio versus angle.

Fabrication of the helical antenna was still underway at the time of demonstration and was not completed.

IV. Low Noise Amplifier

A low noise amplifier was built around a BFP740ESD bipolar transistor. The amplifier was characterized with S-parameters, bias measurement, and a P1dB measurement. The measured results were comparable to simulation.

The high level procedure is as follows: 1. Select DC bias point based on datasheet information. 2. Select topology and optimize based on biasing, gain, and NF needs. 3. Layout the PCB in Altium. 4. Validate results.

1. DC Biasing

From the BJT's datasheet, most of the active regions were contained from Vce = 2-3.5V [6]. At Vce = 3V, min NF occured around Ic = 6mA and max gain was at Ic = 25mA. From here, Ic = 15mA was selected as a compromise for optimal NF and gain. With Ic defined, the current gain, Hfe = Ic/Ib, was estimated to be 190 from figure 4 of the datasheet. Having chosen a 5V supply, the calculation for the bias resistors on the collector is simply: 5 - Rc(15mA) = 3.5V, hence Rc = 100 and two 50 ohm resistors were chosen. Note that splitting Rc in two (instead of one single 100 ohm) allows Vce to be 3.5V instead of 3.85V, which is further from cutoff.

2. Topology

A resistive negative feedback topology is chosen. The choice of resistive feedback is common to make for more stable, broadband designs. In this case, the topology made it so that the amplifier did not need input or output matching, was near unconditional stability, and could be biased with only one power supply. One downside is that the broadband nature of the feedback topology means that low frequencies have huge gain (25 dB) and as a result, measures must be taken to stabilize the amplifier at low frequency.

The choice of inductive emitter degeneration was to push the LNA to unconditional stability (Mu, Mu' > 1) and improve input match. The collector resistance also helped increase stability and output match at the cost of gain and compression point. Figure 10 showcases the huge range of input impedances (circles) for good NF and gain. After optimization the following values were chosen: Rf = 71.5k, Lemitter = 0.12nH, Rcollector = 250. Final simulated specs. are shown

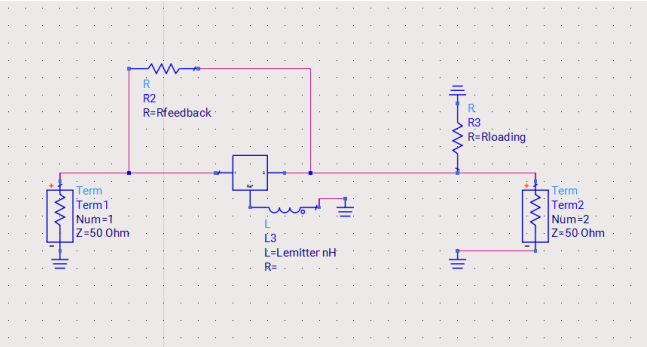


Figure 9. Initial LNA topology (no biasing).

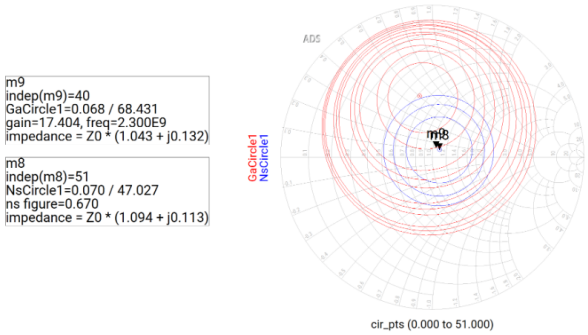


Figure 10. NF (0.7dB) & gain circles (17dB).

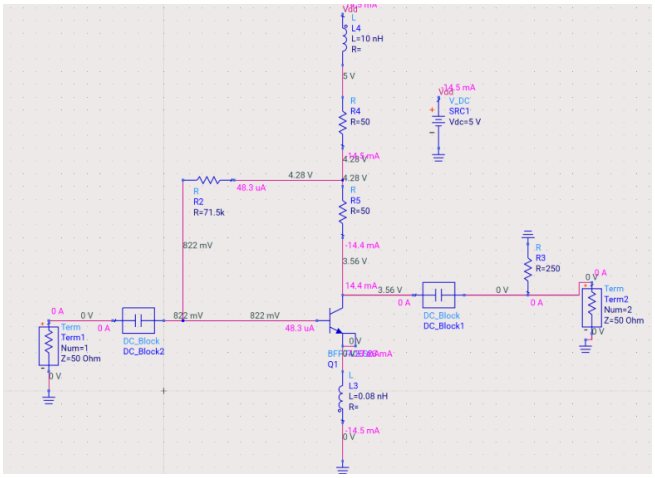


Figure 11. Final ADS workbench.

in figure 14 and 15.

More complicated schemes using L's and C's to kill gain (for stability) only at low frequency were successfully simulated. However, these small improvements in gain and NF at 2.45GHz were not implemented for the sake of simplicity, only 7 total components were used.

3. Altium Layout

To achieve minimal parasitic effects, it is best practice to layout with short traces and minimal discontinuities. In order to have a 50 ohm PCB line comparable in size to the 0402/SOT23 component packages, a 4-layer stackup with coplanar waveguides (CPWG) is implemented. This is due to the fact that small dielectric height corresponds to larger effective wavelengths. Thus, a 4-layer board with h = 0.2mm dielectric thickness allows a 50 ohm trace to be only 11 mils wide, compared to the 50 mils previously needed on a standard 2-layer stackup (h = 1.6mm). CPWG traces also allow for slightly smaller widths and improve coupling at GHz frequencies. Altium's CPWG impedance calculator was used to achieve the correct dimensions for 50 ohm lines as well as the trace inductance (0.12nH) used on the emitter.

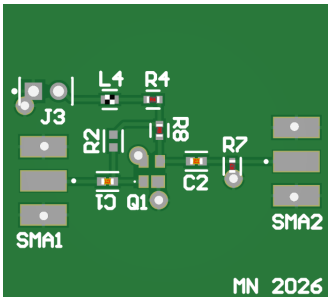


Figure 12. Final LNA layout.

4. Measurement

The vector network analyzer was calibrated from 2.00 to 3.00 GHz in 100 kHz steps using a basic calibration and an Anritsu calibration kit rated to 9 GHz. Cable selection was important during testing. Short cables of about 4 in resting on a steady surface were required. Longer cables of about 18 in produced discrepancies of up to 9 dB on all S-parameters depending on cable movement.

The PCB was supplied with 5 V. S-parameters were measured first. The input power on port 1 was then swept from -40 to 0 dBm using the power and attenuators menu to obtain P1dB. The measured and simulated results are compared in Table 1.

TABLE 1. LNA performance at 2.45 GHz.

Metric	Simulated	Measured	Datasheet Max
S11	-11.8 dB	-5.8 dB	N/A
S22	-17 dB	-9 dB	N/A
S21	18.1 dB	15 dB	20–25 dB
P1dB	-11.6 dBm	-10.65 dBm	N/A
OP1dB	+6.5 dBm	+4.35 dBm	+10.5 dBm

The output P1dB satisfies the relation $\mathrm{OP1dB} = \mathrm{P1dB} + \mathrm{S21}$. The measured S21 of 15 dB is below the datasheet range of 20 to 25 dB. Two adjustments were identified to push the gain higher. The first is to scrape off part of the 0.15 nH emitter inductance trace. The second is to replace the 250 ohm output resistor with a value in the 700 ohm to 2 k range.

The test setup is shown in Fig. 13.

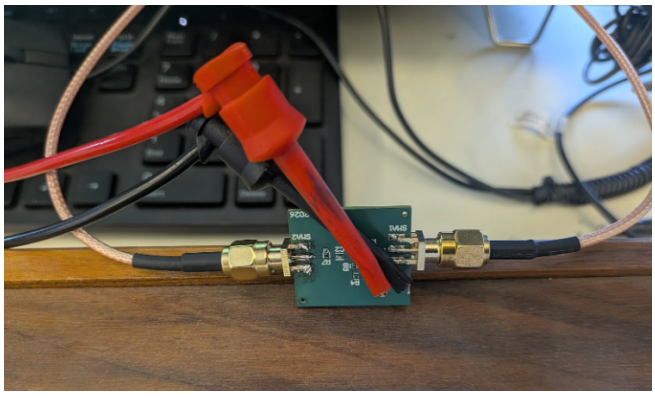


Figure 13. LNA test setup.

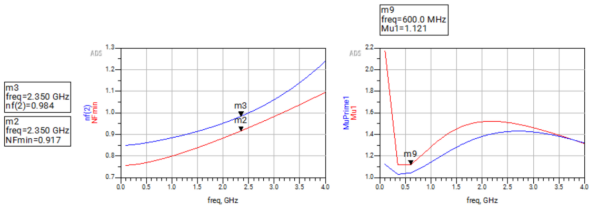


Figure 15. Simulated NF and unconditional stability.

The simulated S-parameters and P1dB are shown in Fig. 14 and Fig. 16.

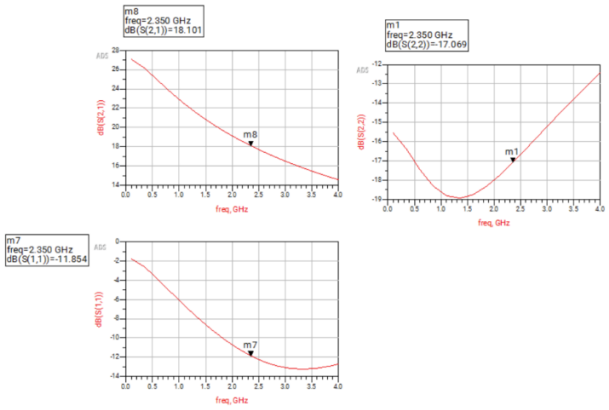


Figure 14. Simulated LNA S-parameters.

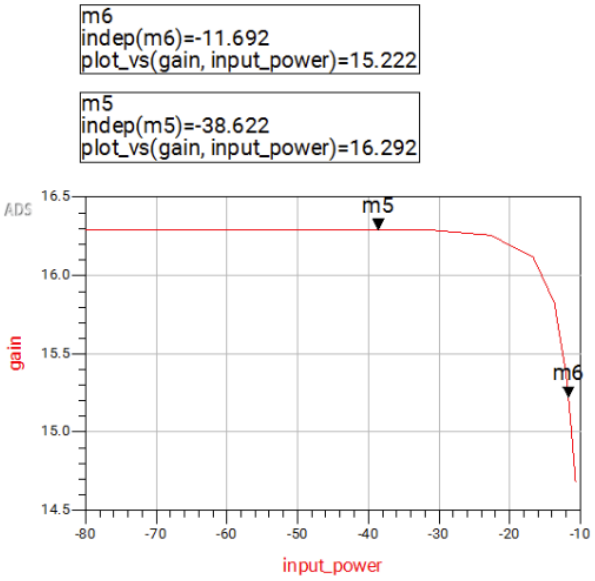


Figure 16. Simulated LNA P1dB.

The measured S11, S22, S21, and P1dB are shown in Fig. 17 through Fig. 20.

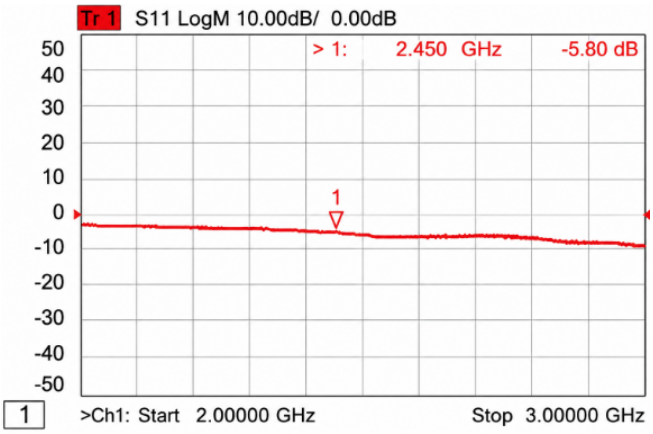


Figure 17. Measured LNA S11.

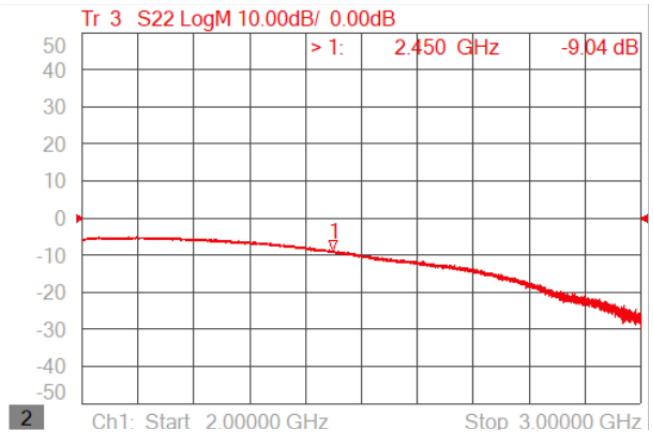


Figure 18. Measured LNA S22.

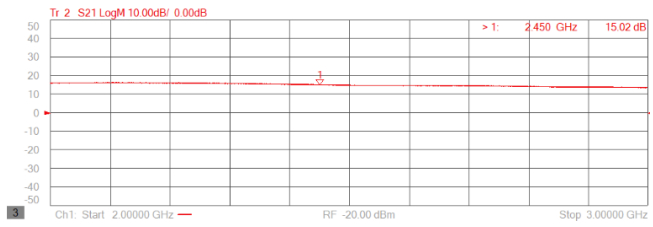


Figure 19. Measured LNA S21.

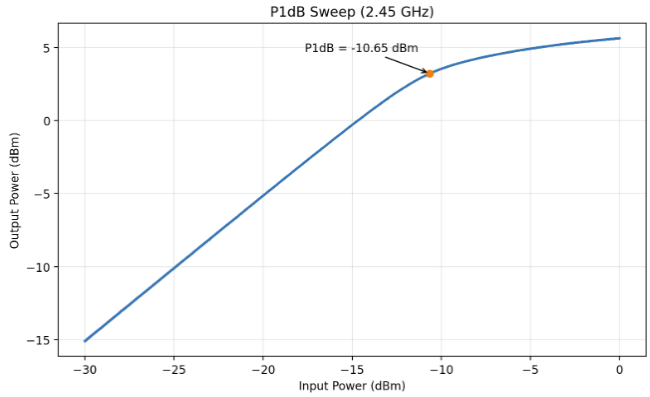


Figure 20. Measured LNA P1dB plotted from the .prn data file.

V. Bandpass Filter

A fifth-order Chebyshev hairpin bandpass filter was designed with a center frequency of 2.45 GHz and a bandwidth of 100 MHz. The filter is based on cascaded coupled half-wavelength resonators. OshPark board properties were used throughout the design.

A. Single-Element Hairpin

Each resonator was first modeled as a straight microstrip trace in ADS using LineCalc. The initial width and length estimates were 2.85 mm and 33.6 mm. With an inner-hairpin line spacing of 5 mm, the initial folded length was set using

$$l_{len} = \frac{\lambda/2 - l_{gap}}{2} = \frac{33.627 - 5}{2} = 14.31 \text{ mm}. \quad (1)$$

A single-element layout was then built and tuned with EM simulation. The initial layout resonated near 4.3 GHz, which is too high and indicates that the electrical length was too short. The length was scaled using the proportional relationship between length and resonant frequency,

$$l_{new} = l_{old}\,\frac{f_{old}}{f_{new}} = 1.85\,l_{old}, \quad (2)$$

which increased the total length to 62.15 mm. After this adjustment the single element resonated near 2.44 GHz with an insertion loss of -3.28 dB.

B. Direct Feed

The input and output feed points were set using the quality factor required by the Chebyshev response. The 0.1 dB ripple coefficients are

$$g_0 = 1,\ g_1 = 1.1468,\ g_2 = 1.3712,\ g_3 = 1.9750,\ g_4 = 1.3712, \quad (3)$$

with $g_5 = 1.1468$ and $g_6 = 1$. The fractional bandwidth and the required external Q are

$$\text{FBW} = \frac{f_h - f_l}{\sqrt{f_l f_h}} = 0.0408, \quad Q_{e1} = Q_{e5} = \frac{g_0 g_1}{\text{FBW}} = 28.09. \quad (4)$$

Because the filter is symmetric, only one feed side needed tuning. The tap distance was adjusted until the external Q measured from the phase response matched the target. Adding

the direct feed line downshifted the resonance from 2.45 GHz to 2.41 GHz, so the resonator electrical length was shortened by 50 mil to recover the design frequency. The resulting external Q was approximately 29.5. The final dimensions were a 2362.2 mil resonator length, an 1100 mil stub length, and a 347 mil feed point.

C. Inter-Hairpin Coupling

The coupling between adjacent resonators was characterized by sweeping the spacing from 4 mm to 6 mm in 0.5 mm increments and recording the two split resonant frequencies at each spacing. The coupling factor was computed from

$$k = \frac{f_2^2 - f_1^2}{f_2^2 + f_1^2}. \tag{5}$$

The results are listed in Table 2. The coupling factor increased approximately linearly with decreasing spacing.

TABLE 2. Hairpin-to-hairpin coupling sweep.

Spacing (mm)	f_1 (GHz)	f_2 (GHz)	k
4.0	2.382	2.468	0.03545
4.5	2.388	2.464	0.03132
5.0	2.392	2.456	0.02640
5.5	2.398	2.452	0.02226
6.0	2.404	2.448	0.01813

The required coupling coefficients were found from the Chebyshev coefficients using

$$M_{i,j} = \frac{\mathrm{FBW}}{\sqrt{g_i g_j}}. \tag{6}$$

This gives $M_{1,2} = M_{4,5} = 0.03255$, which maps to 4.35 mm, and $M_{2,3} = M_{3,4} = 0.0248$, which maps to 5.25 mm.

D. Complete Filter and Optimization

An initial complete filter was constructed from the feed and spacing values above. Its passband insertion loss was about -11 dB, although the center frequency was close at 2.41 GHz. A parameterized layout was then built so that parametric studies could find improved dimensions, which reduced the passband insertion loss to about -5 dB.

The filter was then optimized in Genesys and analyzed in RFPro. The best result was a Chebyshev response with 0.025 dB of ripple, a characteristic impedance of 43 ohm, and a nominal bandwidth of 2.19 to 2.51 GHz. The RFPro analysis gave a passband insertion loss of -4.2 to -4.4 dB across the 2.4 to 2.5 GHz band.

Previous groups observed a fabrication downshift of about 100 MHz relative to the Momentum simulation. The same downshift was assumed, and about 50 MHz of center frequency offset was added to account for it. The target simulated 3 dB bandwidth was roughly 2.37 to 2.65 GHz. The final optimizer settings produced a bandwidth of 2.22 to 2.54 GHz, a characteristic impedance of 42 ohm, 0.025 dB of ripple, and I/O lines of 25 degrees. The passband is wide enough that a downshift of 0 to 100 MHz results in at worst about -5.7 dB of insertion loss across the operating band.

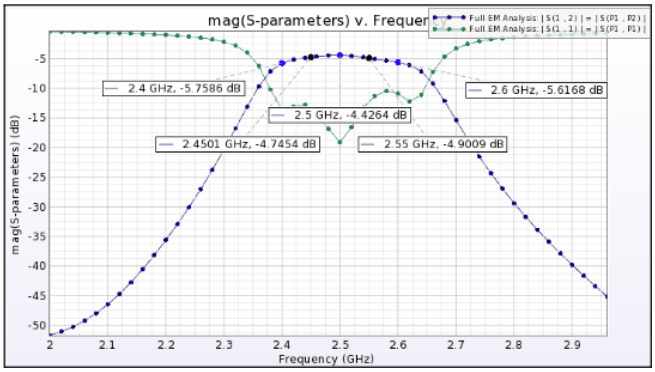


Figure 21. Final hairpin filter layout and EM response.



Figure 22. Final hairpin filter Gerber file.

VI. Modulator and Demodulator

The TRF3722 modulator and TRF3711 demodulator were tested over a hardwired link to find the configurations needed for proper operation. Both devices use accompanying software to set RF center frequency, reference local oscillator frequency, gain, and filtering.

A 1 Vpp, 1 kHz square wave with a 0.5 V DC offset, as recommended by the datasheet, was applied to the in-phase input of the modulator during initial bring-up. DC bias voltages of 6 V and 5 V were applied to the modulator and demodulator. The reference local oscillator frequency was set to 61.4 MHz, which matches the crystal oscillator inside the modulator. A hardwired connection was made between the reference oscillator ports of the modulator and demodulator. All unused ports were terminated with 50 ohm loads to maintain isolation. Only the in-phase component was used to determine the link gains and losses. The recovered in-phase output is shown in Fig. 23.

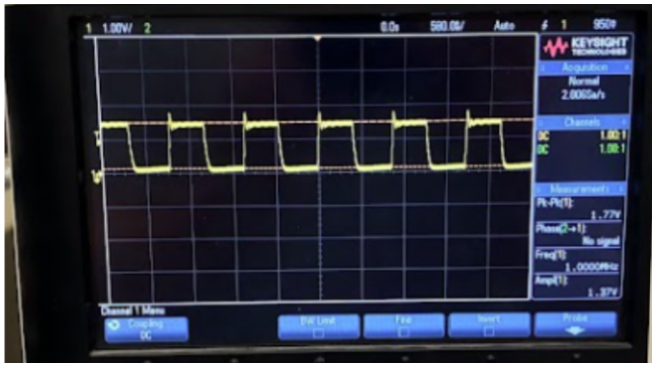


Figure 23. In-phase output of the modulator and demodulator link.

VII. Antenna Measurement Test Stand

A two-axis test stand was built to measure antenna radiation patterns. The firmware runs on an STM32L4A6ZG and is available at `https://github.com/jackm2983/mmwLab`. The stand fits the antenna mount and positions the antenna under test while sampling received power. The coax cable passes through a bearing hole and connects to the LNA. The stepper motor drivers require a 20 V supply.

The stand produces a real-time polar plot while an input is swept. A swept analog input from 0 to 3 V and back to 0 V was used to verify the live plotting path. Additional measurement plots were generated from synthetic test data to confirm the polarization waterfall and radiation pattern processing, including a co-polarization and cross-polarization comparison and a half-power beamwidth readout. The test stand is shown in Fig. 24 and an example polar plot is shown in Fig. 25.

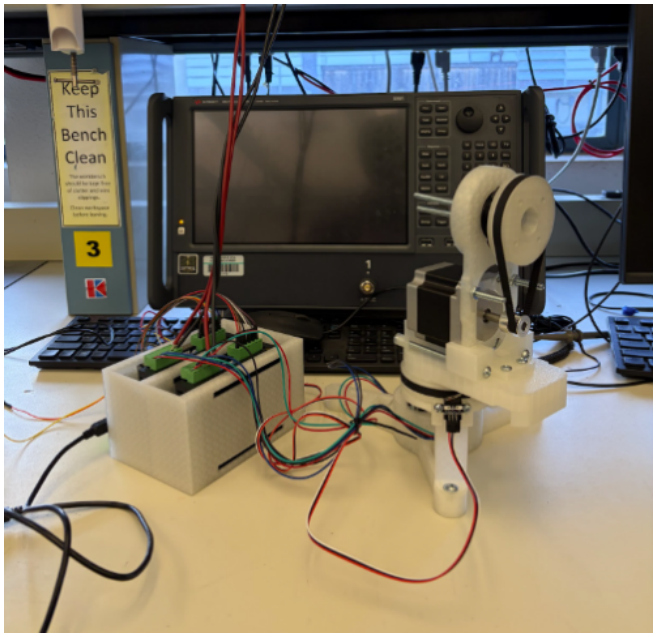


Figure 24. Antenna measurement test stand.

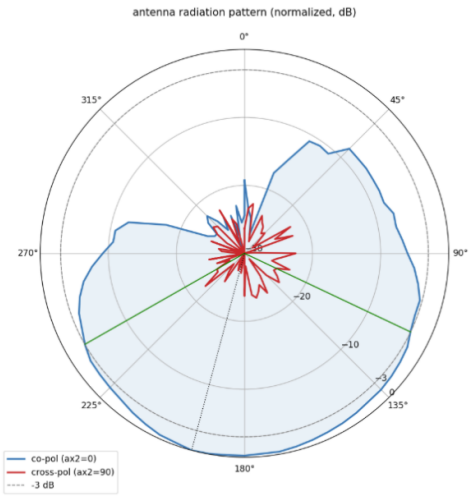


Figure 25. Measured polar plot from swept analog input.

VIII. Link Budget

The far-zone distance was computed to set the minimum separation between the transmit and receive antennas,

$$r_{ff} = \frac{2D^2}{\lambda} = \frac{2(0.241)^2}{0.122} = 0.945 \text{ m}. \tag{7}$$

The free-space path loss at 1 m is

$$L_p = 20 \log_{10}\left(\frac{4\pi df}{c}\right) = 40.2 \text{ dB}. \tag{8}$$

The test range was set to 1 m so that the antenna under test sits in the far zone of the transmit antenna.

The gains and losses of the system components are listed in Table 3. Applying these to the modulator output and stepping through the chain gives the stage powers in Table 4.

TABLE 3. Link component gains and losses.

Role / Device	Gain or Loss (dB)
Power Amplifier	15.8
Tx Bandpass Filter	-4.4
Tx Antenna (Horn)	15
Path Loss (1 m)	-40.2
Rx Antenna (LPDA)	11.8
Low Noise Amplifier	15
Rx Bandpass Filter	-4.4

TABLE 4. Stage power through the link.

Stage	Power (dB)
Modulator Output	-29
Tx Antenna Input	-17.6
Rx Antenna Output	-31
Demodulator Input	-20.4

The modulator output is about 1 mW. With a demodulator input of -20.4 dB and a modulator output of -29 dB, the link

margin is about 8.6 dB. The positive margin indicates that the measurement system has enough signal to operate over the 1 m range.

IX. System Integration and Results

After each subsystem was characterized, the full link was assembled on the bench. The final test setup is shown in Fig. 26.

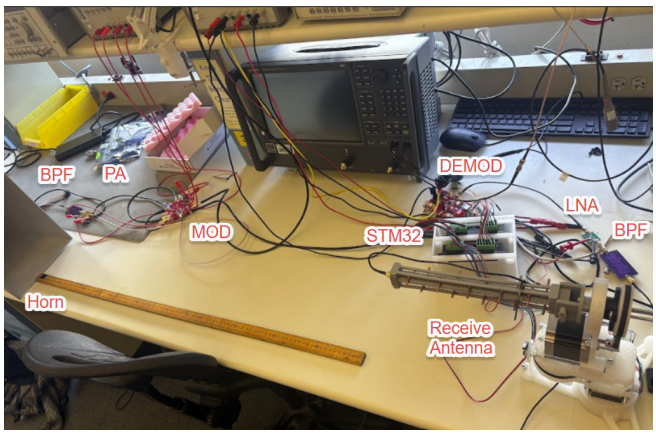


Figure 26. Final integrated test bench for the 2.45 GHz link.

A 40 kHz square wave was applied to the in-phase (I) input of the modulator with the quadrature (Q) input held at zero. The modulated signal passed through the power amplifier, the transmit bandpass filter, and the transmit antenna, propagated across the 1 m range, and was received by the log-periodic dipole array. The received signal was amplified by the low noise amplifier, filtered by the receive bandpass filter, and demodulated by the TRF3711. The recovered in-phase baseband signal was routed to the ADC of an STM32L4A6ZG.

The assembled system demonstrated end-to-end functionality. A baseband signal applied at the modulator produced a corresponding response at the demodulator output after passing through the complete RF chain. This response was observed on an oscilloscope, confirming that the transmit and receive paths, amplification and filtering stages, and antennas operate together as a working link.

However, the ADC capture of the recovered 40 kHz square wave was not accurate and STM32L4A6ZG was unable to characterize the antenna. Possible causes of this error include the behavior of the STM32L4A6ZG ADC itself, the SMA-to-pin connection between the demodulator output and the STM32L4A6ZG input, or another factor in the baseband interface. Further work is needed to determine whether the issue originates in the ADC configuration, the physical connection, or the demodulator output stage.

X. Conclusion

This project designed and integrated a complete 2.45 GHz communication and antenna measurement system. The LPDA reached a simulated boresight gain of 11.8 dBi with about 11 dB of return loss at the design frequency, and the helical antenna reached about 12.3 dBi with a strong match near 2.42 GHz. The low noise amplifier measured 15 dB of gain at 2.45 GHz with a measured P1dB of -10.65 dBm. The hairpin bandpass filter achieved a passband insertion loss of -4.2 to -4.4 dB across the 2.4 to 2.5 GHz band. The modulator and demodulator were configured and verified over a hardwired link, and the two-axis test stand produced live radiation pattern plots.

The link budget gives a positive margin of about 8.6 dB over a 1 m range, and the assembled system showed end-to-end functionality. The remaining issue is accurate baseband capture at the receiver, where the ADC was unable to accurately capture the 40 kHz square wave and the cause was not determined. Future work is to complete the helical antenna fabrication, raise the LNA gain toward the datasheet range, and resolve the ADC issue by testing the STM32L4A6ZG ADC configuration and the SMA-to-pin connection separately.

References

[1] D. M. Pozar, *Microwave Engineering*, 4th ed. Hoboken, NJ: Wiley, 2012.

[2] J.-S. Hong and M. J. Lancaster, *Microstrip Filters for RF/Microwave Applications*. New York: Wiley, 2001.

[3] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed. Hoboken, NJ: Wiley, 2016.

[4] Texas Instruments, “TRF3722 Integrated IQ Modulator with PLL/VCO,” datasheet.

[5] Texas Instruments, “TRF3711 Integrated IQ Demodulator with PLL/VCO,” datasheet.

[6] Infineon Technologies, “BFP740ESD Low Noise Silicon Germanium Bipolar RF Transistor,” datasheet.